# COST AWARE STOPPING BOUNDARIES IN REASONING LANGUAGE MODELS

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## Abstract

Additional reasoning can repair an incorrect answer, replace a correct answer, or consume computation without sufficient improvement. This thesis formulates response-level stopping as a finite-horizon decision based on observable prefixes and hidden correctness. It derives the binary repair-corruption drift identity, applies the standard Bellman/Snell optimal stopping construction, gives a sufficient persistence condition for a drift-sign rule, and constructs exact counterexamples to unconditional myopic optimality. The experimental record separates a variable-horizon matrix from a standardized corpus of 144,440 saved rows, 28,888 trajectories, and 2,948 tasks. Recomputed cluster-based tables show model- and domain-dependent continuation value and matched effects of estimator, token-cap, and precision changes. The historical stacked ROC-AUC of 0.955156 is retained as a retrospective non-nested diagnostic rather than an online performance guarantee. A fitted prefix controller enforces a two-step floor and prevents future generation after stopping. In a newly executed evaluation on 100 GSM8K questions, its actual arm saves 56.51 percent of completion tokens with 7 correct answers versus 6 at the full horizon. On 20 traps it saves 52.11 percent, with one correct answer in each arm. Every learned run stops at step two; adaptive benefit and accuracy noninferiority remain unestablished. The results support protocol-specific, cost-sensitive stopping experiments while identifying limits of myopic rules, probability calibration, and accuracy preservation.



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# Table of contents


Abstract ii

List of tables vi

List of figures vii

Chapter 1 Introduction and motivation 1

1.1 Reasoning as a sequential allocation problem 1

1.2 Operational meaning of overthinking 2

1.3 Research questions 3

1.4 Contributions 4

1.5 Evidence and reproducibility 5

1.6 Related work and the thesis position 5

1.7 Thesis organization 9

Chapter 2 Mathematical formulation and stopping theory 10

2.1 Probability model and admissible information 10

2.2 Exact binary correctness drift 12

2.3 General finite-horizon optimal stopping 14

2.4 When a drift-sign boundary is optimal 16

Exact counterexample 1: delayed repair 17

Exact counterexample 2: a population curve misses adaptive value 18

2.5 Partial observation and feature models 18

2.6 Calibration and perturbation bounds 19

2.7 What sequential uncertainty control can certify 21

2.8 Scope of the empirical claims 24

Chapter 3 Experimental setup and methods 26

3.1 Corpus separation 26

3.2 Model roster 27

3.3 Benchmark tasks and splits 29

# Table of contents (continued)


3.4 Response and revision protocol 30
3.5 Recorded signals and grading 31
3.6 Statistical units and grouping 32
3.7 Outcomes and computation measures 33
3.8 Reproducibility and evidence freeze 34
Chapter 4 Empirical evidence of overthinking 35
4.1 Population transitions and net value 35
4.2 Model and domain differences 37
4.3 Controlled estimator comparisons 38
4.4 Token cap and numerical precision 39
4.5 Ranking and stopping performance 40
4.6 The stacked retrospective diagnostic 42
4.7 Replay and failure interpretation 43
4.8 Empirical conclusions 44
Chapter 5 Online stopping and computation trade-offs 45
5.1 Causal controller contract 45
5.2 Policy choice and limitations 46
5.3 Generation integration 46
5.4 Decision latency 47
5.5 Live paired evaluation 48
5.6 Adversarial questions and traps 50
5.7 Deployable prefix probability model 52
5.8 Actual trained-policy evaluation 54
5.9 Replay and Pareto analysis 57
Chapter 6 Discussion and limitations 60
6.1 Scientific interpretation 60
6.2 Correctness labels and mathematical validity 60

# Table of contents (continued)


6.3 Theory and deployable beliefs 61

6.4 Generalization and adaptation 61

6.5 Costs beyond completion tokens 62

6.6 Adversarial behavior and answer selection 63

6.7 Evidence needed for stronger claims 63

6.8 Conclusion 64

References 65

Appendix A Reproducibility and evidence sources 69

Appendix B Claim scope and mathematical review 71

Appendix C Paired accuracy uncertainty 72

Appendix D Retrospective prediction analyses 73

Appendix E Information flow and delayed repair diagrams 75

# List of tables

Table 1. Mathematical objects and assumptions. 24

Table 2. Standardized five-step corpus. 26

Table 3. Model configurations. 28

Table 4. GSM8K transition panel; 19,500 trajectories and 500 task clusters per row. 35

Table 5. Matched effects. 38

Table 6. Stored causal detectors; common task-held-out development evaluation. 40

Table 7. Mutually exclusive archived-policy loss patterns; 5,735 utility losses among 75,965 trajectories. 44

Table 8. Controller-only latency on 100 saved math-question prefixes. 47

Table 9. Actual paired local generation on 100 GSM8K questions. 48

Table 10. Actual paired local generation on the 20-question trap bank. 50

Table 11. Deployable probability heads on 276 archived held-out tasks; repeated rows are not independent trials. 53

Table 12. Actual trained-policy generation paired with the existing 100-question live baseline. 54

Table 13. Actual trained-policy generation paired with the existing trap-bank live baseline. 55

Table 14. Frozen variants replayed on 1,500 saved trajectories. 57

Table 15. Claim evidence sources. 69

Table 16. Claim scope and required evidence. 71

Table 17. Additional retrospective prediction analyses. 73

# List of figures

Figure 1. Population accuracy and net gain. 36

Figure 2. Micro, domain-macro, and worst-domain ranking differ substantially. 42

Figure 3. Development replay accuracy against completion-token cost. 58

Figure 4. Actual generated arms on two development panels. 59

Figure 5. Offline fitting and runtime information flow. 75

Figure 6. Delayed repair defeats a myopic stopping rule. 76